Evaluating Comparable-Sales and Machine Learning Approaches for House Price Prediction

CSC 542: Statistical Learning with Applications

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Final Submission

Table of Contents:

I. Introduction……...……3

II. State-of-the-Art …….….4

III. Materials and Methods ……..……..…..…5

IV. Evaluation ..……..…..……..…10

V. Result ……..……....……..…10

VI. Conclusion……..……..…17

VII. Discussion & Next Steps……..…18

1. Introduction

The United States housing market has experienced dramatic shifts over the past two decades. Median home prices have increased significantly, reaching approximately $360,591 in February 2026, up 0.2% over the past year (Zillow, 2026), while median household incomes have grown at a much slower rate (U.S. Census Bureau, 2025). This divergence between housing prices and income has created an affordability crisis affecting millions of Americans. The share of first-time buyers has fallen to a record low of 21% (National Association of Realtors [NAR], 2025), existing homeowners face refinancing challenges, and real estate professionals need more sophisticated methods to serve diverse market segments.

In today's market, the dynamics between buyers and sellers have become increasingly complex. A seller's market, which is characterized by low inventory and high demand drives prices up often leaving buyers overpaying for properties. Conversely, in a buyer's market, excess supply gives buyers more negotiating power, but accurately identifying a fair price remains a challenge. According to the U.S. Census Bureau (2025), median household income was $83,730 in 2024, remaining largely flat from the prior year, while home prices have continued to rise, making accurate valuation even more important for buyers, sellers, and lenders. The average 30-year fixed mortgage rate stands at 6.37% as of April 2026 (Freddie Mac, 2026), meaning monthly payments have increased significantly, further straining affordability for prospective buyers.

Accurate home valuation is important for buyers, sellers, real-estate agents, and financial institutions. Traditionally, one of the most common ways to estimate a home's value is the "comparable sales" approach (commonly referred to as "comps"). The idea is simple: find a few recently sold homes that look similar (same area, similar size, similar features), and use their

sale prices to estimate the target home price. That approach works well when the neighborhood is fairly uniform and there are plenty of truly comparable homes. But it gets harder when the market is diverse: different renovation levels, views, lot sizes, and quality differences can make it difficult to decide what “similar” really means.

This project aims to compare the traditional comps approach (represented by KNN regression) against an advanced Machine Learning model (Random Forest regression). We seek to understand not just which is more accurate overall, but how they perform in different price tiers and under conditions of information scarcity. By doing so, we aim to provide insights that help buyers and sellers understand whether a given home is fairly priced in today’s complex market.

To test our hypothesis, we utilize the Ames Housing Dataset (De Cock, 2011). Although the data reflects home sales from 2006 to 2010, it contains 79 distinct attributes that provide a rich and detailed framework for comparing KNN and Random Forest models. This dataset allows us to evaluate how modern machine learning algorithms capture complex property characteristics that traditional comparable-sales methods may overlook.

2. State-of-the-art

Modern real estate valuation has evolved from manual appraisal to Automated Valuation Models (AVMs). Industry leaders like Zillow use their ‘Zestimate’ model, which relies on 7.5 million statistical and machine learning models that analyze hundreds of data points on each property (Kaggle, 2019). While ML models excel at capturing non-linear patterns, the comparable sales logic remains a vital baseline for interpretability, particularly for unique high-end properties where data is sparse.

Prior studies have confirmed that ensemble models like Random Forest (RF) and Gradient Boosting Machine (GBM) consistently achieve lower prediction errors than traditional methods (Ho et al., 2021). Applying these algorithms to over 40,000 housing transactions, the results showed that RF and GBM outperformed Support Vector Machine (SVM) across all three performance metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Notably, both models struggled with extreme-valued properties, showing higher bias at the high end of the price spectrum. This gap in performance for high-end properties is something this study further investigates through price tier analysis.

The K-Nearest Neighbors (KNN) algorithm has been applied to house price prediction as a computational representation of the comparable sales approach, identifying the most similar properties in the training data and averaging their prices (Chen et al., 2026). While KNN is intuitive and interpretable, it is sensitive to noisy or irrelevant features, and its performance can degrade in heterogeneous markets where comparable transactions are sparse (Boehmke & Greenwell, 2020).

Despite advances in ML-based valuation, current models still struggle with luxury and unique properties where comparable transactions are rare. Both RF and GBM have shown higher prediction errors at the high end of the price spectrum, highlighting the need for further investigation into tier-specific model performance (Ho et al., 2021).

3. Materials and Methods

3.1 Dataset

The study uses the Ames Housing Dataset (De Cock, 2011). After removing homes with GrLivArea above 4,000 square feet, the final dataset contains the remaining observations used

for modeling. While the dataset provides 79 explanatory variables, our research focuses on a selection of features to balance model complexity with interpretability. To ensure model robustness, we implemented a strict data filter to exclude properties with a living area (GrLivArea) exceeding 4,000 square feet. Following the dataset author's documentation, these massive properties often represent atypical market outliers that do not adhere to standard pricing logic. Removing them allows our models to better capture the trends of the mainstream residential market. Additionally, categorical missing values (e.g., Pool Quality, Garage Finish) were recoded as "None" to treat the absence of a feature as a deliberate market attribute rather than missing information. The selection of our eight high-impact features is grounded in the Hedonic Pricing Theory (Rosen, 1974), which posits that a property's value is the sum of its individual attributes. We categorized these into four critical "pillars":

- Size: Total Living Area (TotalSF) and land size (LotArea).
- Quality: Overall structural grade (OverallQual) and kitchen renovation level (KitchenQualNum).
- Age: The modernity of the home (HouseAge), calculated by the difference between the year sold and year built.
- Utility & Location: Total bathroom count (TotalBath), garage capacity (GarageCars), and a neighborhood-based price proxy (NB_MeanLogPrice).

A key strength of our study is the Feature Engineering process. We created the `TotalBath` metric by combining full and half baths from all levels, reflecting real-world buyer behavior where total utility outweighs specific floor placement. Furthermore, we "mathematized" location by creating `NB_MeanLogPrice`, converting abstract neighborhood names into a historical price index to capture the "location premium." Lastly, ordinal qualities (e.g., Kitchen Quality) were mapped to a numeric 0–5 scale (Label Encoding) to allow for mathematical weighting. As shown

in Figure 1, the original `SalePrice` distribution was heavily right-skewed, which can lead to heteroscedasticity—where a model's errors grow disproportionately with the price.

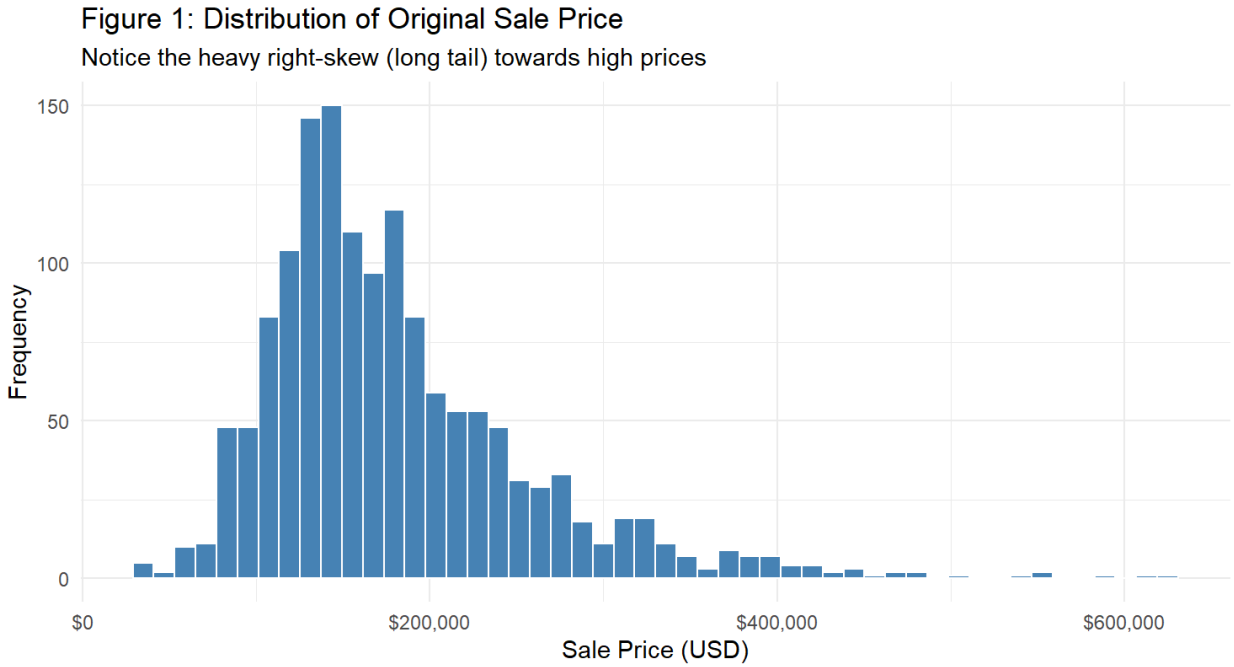


Figure1: Distribution of Original Sale Price

The original SalePrice distribution is heavily right-skewed, with most homes priced between $100,000 and $200,000 and a long tail extending beyond $600,000. This skewness motivated the use of a log-transformation to stabilize variance before modeling.

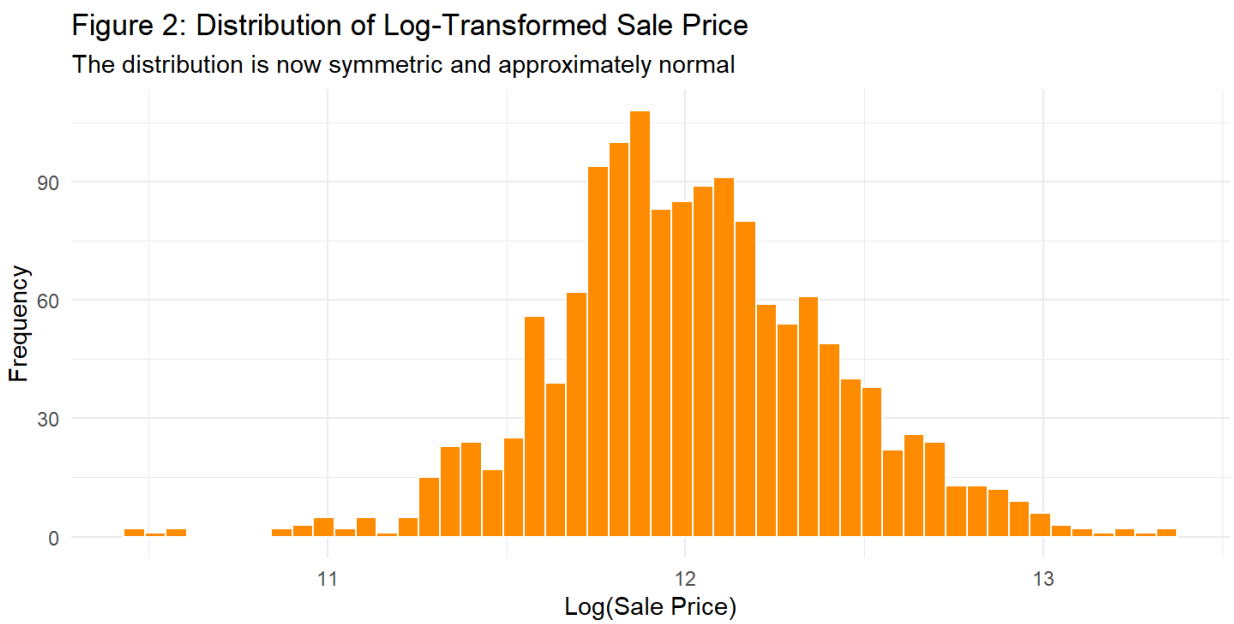
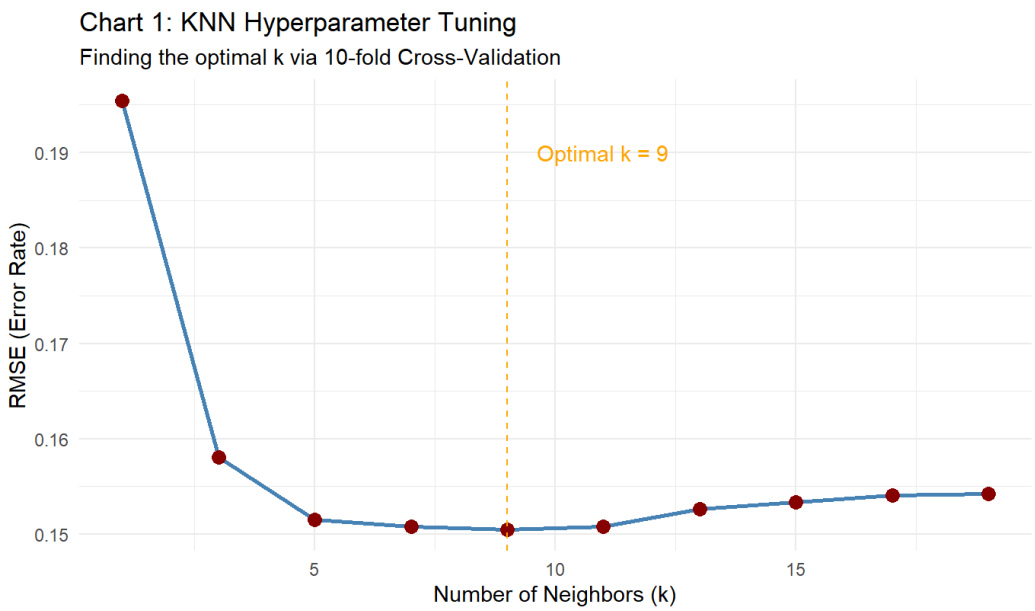


Figure2: Distribution of Log -Transformed Sale Price

3.2 Methods

We selected eight core features for the comparative analysis to represent the primary drivers of real estate value: TotalSF, OverallQual, TotalBath, HouseAge, LotArea, GarageCars, KitchenQualNum, and the engineered NB_MeanLogPrice. To mimic the traditional comparable-sales logic, we implemented K-Nearest Neighbors (KNN) regression using the caret framework. This model estimates a home's value by calculating the Euclidean distance between the target property and its nearest neighbors in a multi-dimensional feature space that includes size, age, quality, land size, and location-related variables. Rather than selecting the number of neighbors arbitrarily, we used 10-fold Cross-Validation to evaluate odd values from k = 1 to 19. This process allowed the model to identify the number of comparable homes that produced the lowest prediction error for the Ames market. As shown in Chart 1, the optimal value was k = 9, where cross-validated RMSE reached its minimum. Because KNN is distance-based, variables measured on larger scales (such as LotArea) could otherwise dominate the similarity

calculation. Therefore, all predictors were standardized using Z-score normalization. To avoid data leakage, scaling parameters were calculated using the training set only and then applied to the test set, ensuring an unbiased model evaluation.



As a contrast to the local similarity logic of KNN, we implemented a Random Forest regression model using the randomForest package. Random Forest is an ensemble learning method that combines the predictions of many decision trees. In this study, the model was trained with 500 trees (ntree = 500). This “Jury of Experts” approach is designed to capture complex, non-linear relationships and interactions among housing features that simple distance-based methods may overlook. For example, the model can learn that a premium kitchen may add more value in a newer home than in an older distressed property. The model used the package default mtry setting for regression, which considers approximately one-third of predictors at each split. With eight predictors in this study, this corresponds to two candidate variables per split. Trees were allowed to grow under the package’s default settings. Unlike KNN, Random Forest is relatively insensitive to feature scaling and does not strictly require normalization. Although the same

scaled feature matrix was used for consistency, we also maintained the same feature set to ensure a fair head-to-head comparison.

4. Evaluation

To ensure a robust and fair comparison between the traditional comparable-sales logic and the machine learning approach, we used the following evaluation framework: the dataset was divided into an 80% training set and a 20% test set. This provides an out-of-sample evaluation of how well each model generalizes to unseen listings. Within the training set, we applied 10-fold Cross-Validation to identify the optimal value of k for the KNN model. This ensured that the comparable-sales baseline was properly tuned before final comparison. Models were evaluated using three complementary metrics:

- Log-RMSE: Measures predictive precision while penalizing larger errors.
- MAE (Mean Absolute Error): Measures the average dollar difference between predicted and actual prices.
- MAPE (Mean Absolute Percentage Error): Measures relative error in percentage terms, which is intuitive for buyers and sellers.

To examine performance across market segments, results were further analyzed within three quantile-based price tiers: Low-End, Mid-Range, and Luxury. This helps identify where automated valuation performs well and where prediction uncertainty becomes greater.

5. Results

5.1 Overall Model Performance

This section evaluates the overall performance of the Random Forest (RF) and k-Nearest Neighbors (KNN) models by comparing actual sale prices with predicted values.

Referencing Chart 2 (Predicted vs. Actual Sale Price), the scatter plot shows how closely the predictions cluster around the dashed 45-degree line, which represents a perfect prediction. Both models demonstrate high alignment in the $100,000 to $300,000 price range, where the majority of the data points are concentrated. This indicates that the core features effectively capture the primary drivers of house value. Moreover, the teal points (Random Forest) are more

tightly clustered around the diagonal line than the coral points (KNN). This confirms that Random Forest, through its ensemble of 500 decision trees, better explains the price variance and offers higher predictive accuracy.

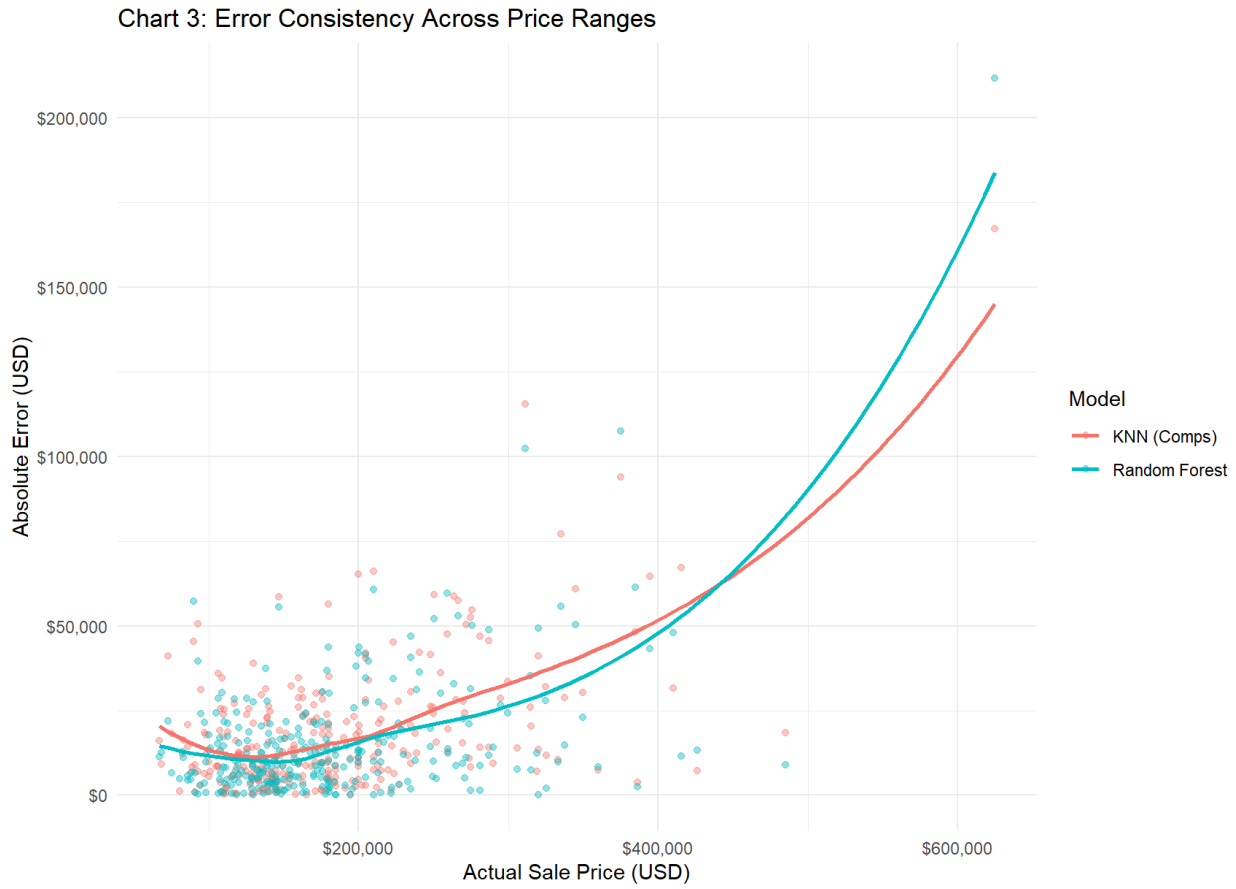


Chart 3 (Error Consistency Across Price Ranges) uses LOESS smoothing lines to visualize how the absolute error in USD changes as the house price increases. As shown in the chart, the error for both models increases significantly once the actual price exceeds $450,000. This reflects the concept of heteroscedasticity—luxury homes are harder to value because they are more unique and less standardized than mid-market homes. Between $100,000 and $450,000,

Random Forest generally shows lower or comparable error than KNN across much of the mainstream market. This suggests RF is more robust in typical pricing ranges.

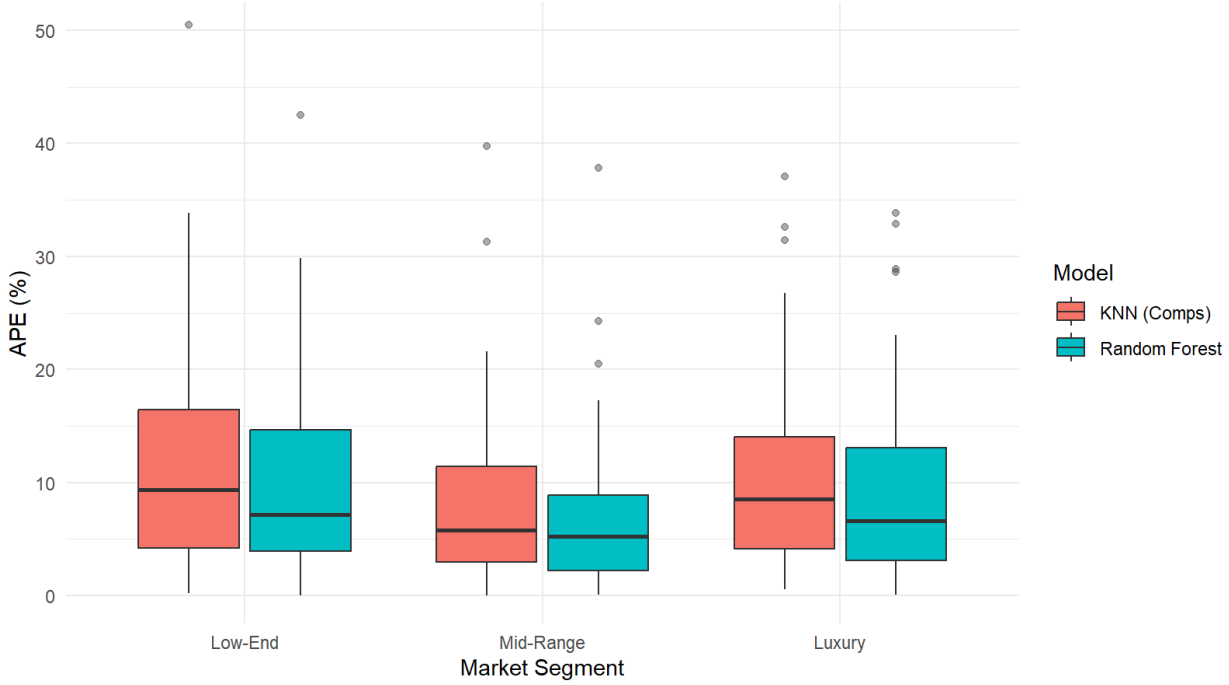
5.2 Segmented Performance & Error Stability Analysis

To better understand the practical reliability of the models, we divided the housing market into three tiers: Low-End, Mid-Range, and Luxury. This helps move beyond overall averages and identify where each model performs best. Referencing Chart 4, the Mean Absolute Percentage Error (MAPE) shows a clear U-shaped pattern.

Moreover, both models achieved their best performance in the Mid-Range segment. Random

Forest reached the lowest error at 6.5%, compared with KNN's 8.0%. This suggests mid-market homes have more standardized features and are easier to predict. Additionally, accuracy declines in both Low-End (RF: 10.3%) and Luxury (RF: 8.9%) tiers. For low-end homes, even small dollar errors create larger percentage errors. For luxury homes, unique premium features make valuation more difficult. While MAPE tells us the average error, Chart 5 (Error Stability) illustrates the consistency and risk profile of each prediction.

Chart 5: Error Stability by Price Tier

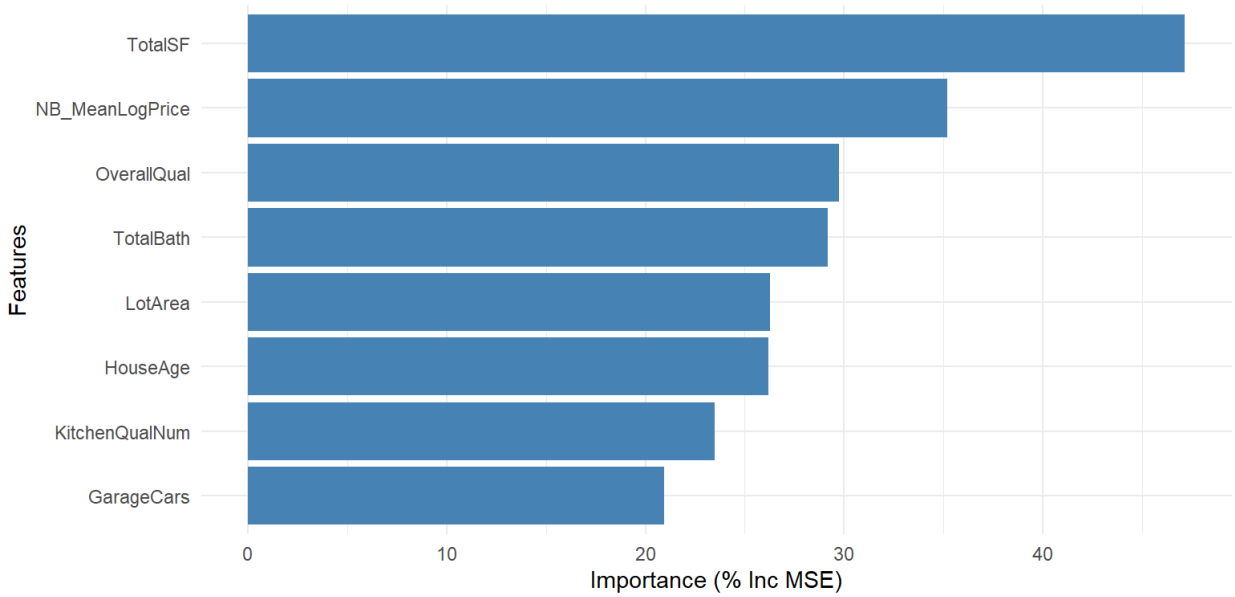


While MAPE shows the average error, Chart 5 focuses on error stability and risk distribution. In addition, across most price tiers, Random Forest displays a slightly narrower interquartile range than KNN, indicating more concentrated errors and lower variability. This suggests RF provides more stable performance across different housing segments. Furthermore, both models generate outliers, but KNN tends to show more extreme upper-tail errors in some segments, especially in the Low-End and Luxury markets. This suggests that a comparable-sales approach may struggle when homes have unusual characteristics, while Random Forest can reduce some extreme errors by combining predictions from many decision trees.

5.3 Feature Importance Analysis

While predictive accuracy is essential, understanding *why* a model assigns a specific value is crucial for real estate stakeholders. Chart 6 illustrates the relative importance of each feature in the Random Forest model, measured by the percentage increase in Mean Squared Error (%IncMSE) when a feature's data is permuted.

Chart 6: Feature Importance (Key Price Drivers)

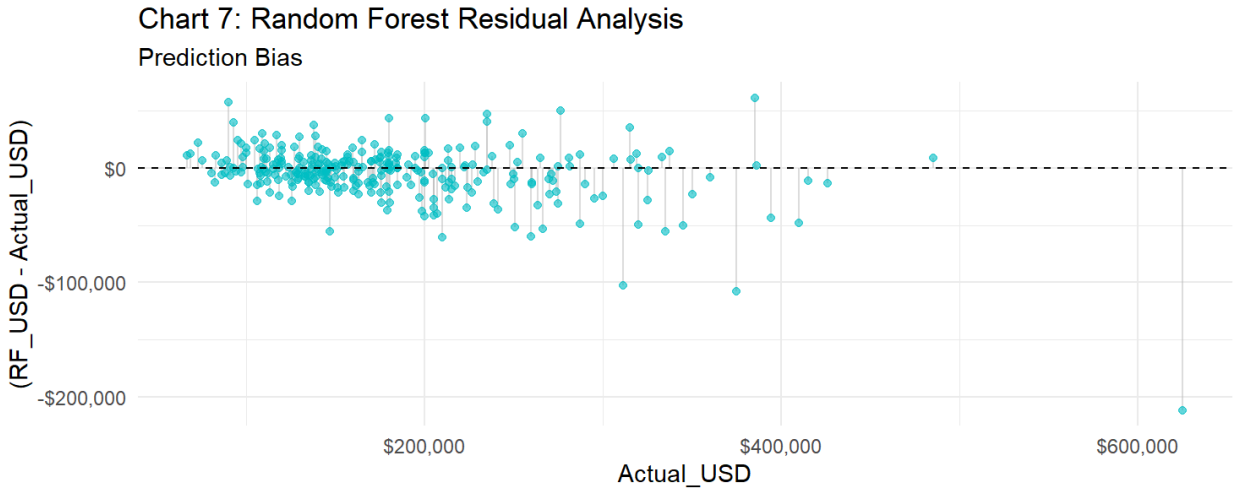


- TotalSF (Total Square Footage) is the most influential variable, confirming the basic real estate principle that property size strongly drives price.
- NB_MeanLogPrice, our engineered neighborhood price index, ranks second. This strongly supports the idea that location remains one of the most important determinants of housing value.
- OverallQual ranks third, indicating that construction quality, materials, and finish level are major pricing factors.

- TotalBath also ranks highly, suggesting buyers place clear value on convenience and usable living space.
- HouseAge likely captures depreciation, maintenance expectations, and demand for newer homes.
- KitchenQualNum indicates that kitchen condition and finish quality are meaningful value signals in the housing market.

5.4 Residual and Bias Analysis

To evaluate the reliability of the Random Forest model, we conducted a residual analysis. Chart 7 plots residuals against actual sale price. A well-calibrated model should display residuals randomly scattered around the zero line.



Moreover, most residuals are distributed around the dashed zero line, suggesting no strong systematic tendency to consistently overprice or underprice homes. However, higher-priced homes show a mild tendency toward negative residuals, indicating occasional underestimation. This suggests that the data transformation and modeling process helped reduce overall bias. In

addition, as sale price increases, the spread of residuals becomes wider, creating a funnel-shaped pattern. This is a common sign of heteroscedasticity. The larger residuals among homes above $400,000 suggest that high-end properties contain more unique characteristics that are harder for a general model to capture, such as custom upgrades, premium locations, or unusual layouts.

6. Conclusion

In conclusion, this study finds that while traditional appraisal logic, represented by the K-Nearest Neighbors (KNN) model, remains a useful and interpretable baseline, the Random Forest (RF) model generally provides a more accurate and robust solution for Automated Valuation Models (AVMs). By combining structural housing characteristics with an engineered neighborhood price index, the RF model achieved lower prediction error across all market tiers, with its strongest performance in the Mid-Range segment where MAPE reached 6.5%. However, as shown in Chart 3 and Chart 7, prediction uncertainty increases substantially in the Luxury tier. This suggests that high-end homes contain more unique features, such as custom finishes, premium locations, and rare layouts which are harder to capture using structured data alone. For industry practice, these findings support a human-in-the-loop strategy: machine learning models can efficiently price standard homes at scale, while human expertise remains valuable for complex or high-value properties. Overall, this project demonstrates how modern data science can complement, rather than replace, traditional real estate valuation methods.

7. Discussion & Next Steps

Several extensions could further improve valuation accuracy. First of all, future models could use computer vision techniques to analyze listing photos and measure visual factors such as curb appeal, renovation quality, and interior design. Moreover, adding mortgage rates, employment trends, school quality, and neighborhood demographics may improve responsiveness to changing market conditions. Gradient boosting methods such as XGBoost or LightGBM may outperform Random Forest by focusing on residual errors. More advanced geographic models could capture distance to parks, downtown areas, transit hubs, and other local amenities beyond simple neighborhood averages.

Link to GitHub repository : https://github.com/XimenaChen/CSC542-Final-Project.git

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